

HR Dashboard – Power BI Project

Overview

This project was completed as part of the **Digital Egypt Pioneers Initiative (DEPI)**. I was given a raw HR dataset (split across three Excel sheets) and tasked with cleaning, modeling, and visualizing it in Power BI to answer key workforce questions – headcount, turnover, compensation, and data availability across departments.

Tools Used

- **Power Query** – data cleaning and transformation
- **Power BI** – data modeling (relationships) and dashboard visualization

Data Source & Structure

The raw data arrived as three separate sheets, which were kept as three related tables rather than merged into one flat table:

Table	Key Fields	Role
Employees	ID, City, Salary, Performance Review, Overdue Vacation?, Last Promotion Date	Employee-level attributes
Sheet1	ID, Department, Employment Status, Gender, Hire Date, Education	Core employee record
Department Bridge	Department, Manager	Department → Manager mapping

Relationships: Employees ↔ Sheet1 on ID ,
and Sheet1 ↔ Department Bridge on

Department .

The Department Bridge table was kept separate instead of repeating the manager's name on every employee row. This avoids data redundancy (normalization) — if a manager changes, it's updated in one place instead of across every row belonging to that department.

Data Cleaning & Transformation (Power Query)

- **Changed Data Types** — corrected columns (dates, whole numbers, text) so calculations and relationships work correctly.
- **Trimmed Text** and **Capitalized Each Word** — applied to text columns (Employee, Department, Education, Position, etc.) to remove inconsistent spacing and casing from manual data entry. Without this, values like "Sales" and "Sales " would be treated as two different categories, silently breaking relationships and undercounting groups in the dashboard.
- **Nulls handled by context, not blanket-removed:**

- Last Promotion Date nulls were **kept** — a blank here reflects a real event (a promotion) that simply hasn't happened yet, not a data problem.
- Availability = Unknown (~20% of records) was **flagged, not ignored** — this reflects a genuine data collection gap rather than a real employee status (see Key Insights).



Dashboard Features

- Filters: Department, Employment Status, Education
- KPIs: Total, Active, and Terminated Employees
- Headcount by Department | Terminations by Department
- Average Salary by Department
- Overdue Vacation Tracker
- Employee Availability Status
- Employee Details Breakdown table



Key Insights

1. Raw counts are misleading for turnover – rates tell the real story.

Development has the highest number of terminations (11), which looks like the biggest problem at first glance. But Development is also the largest department (47 employees).

Calculating the termination *rate* (terminations ÷ headcount) changes the picture: Sales (50%) and Strategy (44%) have proportionally higher turnover than Development (23%). Legal shows 100%, but with only 1 employee total, that figure isn't statistically reliable and shouldn't be treated as a pattern.

2. Averages can hide small sample sizes.

Marketing has the highest average salary (4,314), but it's based on only 4 employees – compared to 47 in Development. A small sample like this is more easily skewed by one or two individual salaries, so this figure should be read as a directional signal, not a reliable department-wide benchmark, without checking the underlying salary distribution.

3. The "Unknown" availability status is a data quality issue, not an employee issue.

~20% of employee records have an Unknown

availability status — this reflects missing data at the recording stage, not an actual employee state. Breaking it down by department shows it isn't evenly spread: Development accounts for 12 of the 18 Unknown records (25.5% of its own headcount) — the highest rate of any department, even after adjusting for its larger size. This points to a specific data collection gap in how Development's records are maintained, worth investigating directly with HR.



What I'd Improve Next

- Investigate *why* Development has both the highest Unknown-data rate and largest headcount — is it a process gap specific to that department?
- Break down the salary averages by individual distribution (not just the mean) to confirm the Marketing figure isn't driven by one outlier.
- Add a calculated Termination Rate measure directly into the dashboard instead of only raw counts, so the more accurate insight is visible at a glance.